Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Object

Meaning in my words In a force diagram, what do we mean by the object?	**How can you tell?** You circle only the cart. Which force belongs: the hand on the cart or the cart on the hand?
Example or sketch What would you put at the centre of a free-body diagram for a hanging lamp?	**Non-example and why** A student switches to forces on the ceiling halfway through. How should they organize those?

External force

Meaning in my words How would you explain external force to a friend?	**How can you tell?** Draw a boundary around a lamp alone. Why is the cable's pull external?
Example or sketch If your system contains two carts and a hand pushes one from outside, what force would you include?	**Non-example and why** If lamp and cable are both inside the boundary, should their mutual pull count as an external force?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Force-arrow scale

Meaning in my words	How can you tell?
How would you explain force-arrow scale to a friend?	Your scale is 1 cm for 2 N. How long should a 6 N arrow be?
Example or sketch	**Non-example and why**
Draw 2 N left and 6 N right at that scale.	Two arrows of equal length are labeled 2 N and 8 N on one diagram. What would you change?

Net force

Meaning in my words	How can you tell?
How would you explain net force to a friend?	Two people push one cart in opposite directions. Why do the arrow directions matter?
Example or sketch	**Non-example and why**
One person pushes 6 N right and another 4 N left. What single arrow replaces both?	A student adds those pushes to get 10 N right. Which push did they count the wrong way?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch the lamp, its boundary, and its two force arrows. Make their lengths predict upward acceleration.

Chosen object: _____

My marks mean

Build the force overlay

15

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Keep only forces on the chosen object in its diagram.

CHOOSE ONE JOB TOGETHER

SYSTEMS ENGINEER ☐

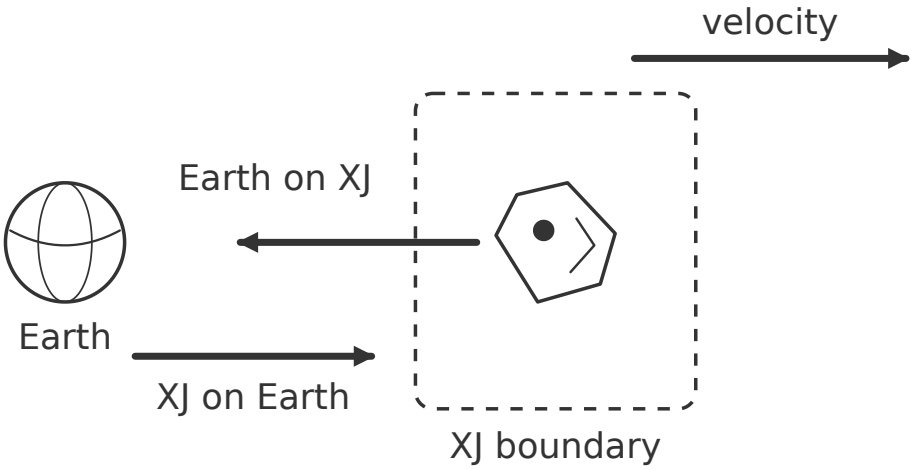
Outline XJ as the chosen object. Keep only the force arrows you would use to calculate XJ's acceleration.

ANIMATION DIRECTOR ☐

Separate the force and motion layers. Tell the animator which Earth arrow belongs in a different shot.

Point to a clue. Keep your first marks; add the update using the color key.

The mission simulator needs one overlay called “forces on XJ.” Its proposed layers include Earth on XJ, XJ on Earth and XJ’s velocity.



01 OUR FIRST CALL

Which arrows belong in 'forces on XJ'? Explain what you keep and move.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Where does the new arrow belong? Update the force layer and explain.

Switch speakers. Must XJ move in the direction of its net force? Give a possible motion that shows your reasoning.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks